

³²P - Comments on evaluation of the decay data

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This evaluation has been carried out with the literature available by February 2026. The former evaluation was completed in 2004 (2004BEZR). Datasets have been averaged with the LWEIGHT (*Limitation of Relative Statistical Weight*) program.

Phosphorus-32 was one of the first artificial radionuclides produced by neutron activation (1934FE01) and, as a typical pure β^- emitters, has been studied regularly since then.

1. Nuclear data

Phosphorus-32 disintegrates by β^- emission (100%) to the ground state of Sulfur-32, as the first excited state of ³²S is above the Q-value of ³²P. The Q value (1710.661 (40) keV) has been adopted from the latest Atomic Mass Evaluation AME2020 (2021WA16). The spins and parities of the ground states of the two nuclei are well known and have been adopted from (1958ST50): 1⁺ for ³²P and 0⁺ for ³²S.

1.1. Half-life

The experimental values (in days) are given in

Table 1 that contains 28 initial values. From these, two are without uncertainties (1935AM01 and 1977BE59) and consequently omitted from any calculation. Two other series come from the same laboratories, so only the last value is retained for each of them: 1937NE02 updated 1936SAGA and 2020UNTE superseded 1994CO02, 2002UN02 and 2014UN01.

The initial average calculation on the 22 remaining independent data using LWEIGHT resulted in a large reduced- χ^2 (27.12). Indeed, in the very first measurements, the presence of ³³P, simultaneously produced in the ³²S(n, γ)³²P activation reaction induced erroneous values. These (1934AM01, 1935PR01 and 1936SI01) were excluded according to the Chauvenet criterion. Several uncertainties were enlarged to take into account details in the references.

In the next run, seven more data were excluded, either due to large uncertainties (1935FA01, 1937NE02, 1938CA02) or to significant discrepant value (1940MU01, 1951SI25, 1953LO10 and 1959RO61). The- χ^2 value was reduced to 2.44, but this set was still discrepant.

The value of 1958DA10 was excluded because its high uncertainty does not carry significant weight in the average. The dataset is inconsistent due to the value of 1957AN03, which is lower than the others and has a too low uncertainty. The dataset becomes consistent by increasing its uncertainty from 0.015 to 0.019. Furthermore, due to the LWEIGHT calculation rule, the uncertainty of 2020UNTE was increased so that its contribution to the weighting would not exceed 50%.

Finally, the selected eleven values (in bold in Table 1) have been combined, giving a weighted average of 14.273 (7) d, with an internal uncertainty of 0.005 d, an external uncertainty of 0.007 d and a reduced- χ^2 of 2.26 which can be compared to the critical χ^2 of 2.32. **The adopted value is 14.273 (7) d.**

Table 1 – ³²P half-life experimental values.

Reference	T _{1/2} (d)	Comment
1934AM01	17.5 (11)	Excluded by the Chauvenet criterion
1935AM01	14	Excluded, no uncertainty
1935FA01	14(1)	Omitted from analysis
1935PR01	15.0 (15)	Excluded by the Chauvenet criterion
1936SAGA	14.0 (3)	Superseded by 1937NE02
1936SI01	15.0 (1)	Excluded by the Chauvenet criterion
1937NE02	14.5 (3)	Omitted from analysis
1938CA01	14.295 (29)	
1938CA02	14.5 (3)	Omitted from analysis
1940MU01	14.07 (1)	Omitted from analysis
1948KL28	14.35 (5)	
1950BA94	14.300 (57)	
1951SI25	14.60 (5)	Omitted from analysis
1953LO10	14.50 (4)	Omitted from analysis
1957AN03	14.223 (19)	Published uncertainty (0.015) increased
1958DA10	14.2 (3)	Omitted from analysis
1959RO61	14.55 (6)	Omitted from analysis
1961MA46	14.282 (10)	
1966GO16	14.290 (14)	
1969LA34	14.268 (42)	
1969PE04	14.32 (2)	
1977BE59	14.2	Excluded, no uncertainty
1977MU07	14.34 (5)	
1979PR14	14.28 (4)	
1994CO02	14.26 (1)	Superseded by 2020UNTE
2002UN02	14.263 (3)	Updated in 2014UN01 and 2020UNTE
2014UN01	14.263 (35)	Superseded by 2020UNTE
2020UNTE	14.262 (3)	Weighting limited to 50%
Adopted	14.273 (7)	Weighted average

1.2. Beta transition

There is only a single β^- transition in the ³²P decay scheme that directly populates the ground state of ³²S; this ground-state-to-ground-state transition is of allowed nature. The transition energy corresponds to the adopted $Q_\beta = 1710.661$ (40) keV.

Thirty experimental values of the maximum energy of the β^- continuous spectrum were examined, as presented in Table 2. Among these, four are without uncertainties (1934AM01, 1937LY08, 1957PO36 and 1960GESH). Several other values come from laboratories that carried out multiple measurements; consequently, only the most recent value has been retained for each of them: 1937NE02, 1951SH92, 1958JO33, 1961NI02 and 1961FE10 respectively replaced 1937LY08, 1950AG01, 1949LA06, 1952JE12 and 1956PO07, and 1958DA10.

The weighted mean of the β^- end-point energy has been calculated using LWEIGHT with 21 values. This first set was discrepant with a χ^2 value of 4.76 and a critical χ^2 of 1.93. The values from 1937NE02 and 1938CA02 were discarded according to the Chauvenet criterion.

To get a consistent set of data, the three values with energy lower than 1.7 MeV (1951SH92, 1957RI41 and 1966CA10) have been excluded. In addition, the uncertainty of 1961NI02, which seems very low compared to other values obtained in the same conditions, has been increased from 1 keV to 2 keV.

Table 2 – Experimental values of the end-point of the β^- spectrum of ³²P.

Reference	E _{max} (MeV)	Comment
1934AM01	1.62	Excluded (no uncertainty)
1937LY08	1.69	Superseded by 1937NE02 - Excluded (no uncertainty)
1937NE02	1.59 (5)	Excluded by the Chauvenet criterion
1938CA02	1.68 (5)	Excluded by the Chauvenet criterion
1946SI09	1.712 (8)	
1949LA06	1.689 (10)	Superseded by 1958JO33
1950AG01	1.718 (10)	Superseded by 1951SH92- Excluded (Chauvenet criterion)
1950WA76	1.708 (8)	
1951SH92	1.695 (5)	Excluded (too low value)
1952JE12	1.704 (8)	Superseded by 1956PO07 and 1961NI02
1954ANTO	1.712 (8)	
1956PO07	1.712 (5)	Superseded by 1961NI02
1957PO36	1.711	Excluded (no uncertainty)
1957RI41	1.695 (15)	Excluded (too low value)
1958DA10	1.705 (4)	Superseded by 1961FE10
1958JO33	1.709 (5)	
1960GESH	1.714	Excluded (no uncertainty)
1961DE25	1.704 (5)	
1961FE10	1.705 (4)	
1961NI02	1.7046 (20)	Published uncertainty (0.001) doubled
1963BO31	1.706 (11)	
1966CA10	1.697 (5)	Excluded (too low value)
1968FI04	1.7101 (10)	
1969FL02	1.7012 (40)	
1971BO01	1.706 (4)	
1971PE03	1.707 (4)	
1971ZE04	1.711 (2)	
1976MO12	1.7120 (8)	
1993GR17	1.710 (3)	
2001KO07	1.708 (2)	
Calculated	1.710 (7)	Weighted average with external uncertainty

The averaging of the sixteen remaining values resulted in a reduced χ^2 of 1.64, compared to a critical χ^2 of 2.04, with an internal uncertainty of 0.5 keV, an external uncertainty of 0.7 keV. Adopting the latter, the **calculated β^- end-point energy is 1710.0 (7) keV**, consistent with the adopted Q-value.

The log $ft = 7.90673$ (22) value have been calculated by means of the BetaShape code version 2.4 (2023MO21). This program also provides an **average spectrum energy of 693.306 (18) keV** from theory, which has been adopted, and 689.484 (18) keV from the experimental shape factor.

2. Atomic data

Fluorescence yield data, relative K X-ray emission probabilities and ratios $P(KLX)/P(KLL)$ and $P(KXY)/P(KLL)$ are from Schönfeld *et al.* (1996SC06). Table 3 summarizes the X-ray emission energies and relative intensities of K X rays of the daughter element (S).

Table 3 – X-ray emission energies of sulphur.

Origin	Energy (keV)	Relative probability (%)
XL (S)	0.1513 – 0.2272	
XK _{α2} (S)	2.30667	50.53
XK _{α1} (S)	2.30787	100
XK _{β1} (S)	2.4641	9.33

3. Production modes

- ³¹P (n, γ) ³²P Possible impurity: ³³P
- ³²S (n, p) ³²P Possible impurities: ³³P, ³⁵S
- ³⁵Cl (n, α) ³²P Possible impurities: ³⁶Cl, ³⁴P

Other modes: ²⁹Si (α, p) ³²P ; ³⁰Si (³He, p) ³²P ; ³¹P (d, p) ³²P ; ³⁴S (d, α) ³²P.

4. Recommendation of measurements

The amount of data available to determine the half-life of ³²P has been extremely limited for more than 40 years, with the latest measurements having been carried out by the same institute. Independent careful measurements, taking into account the improved reliability of today's facilities, would be welcome. Precise measurement of the beta spectrum could also be valuable.

5. References

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